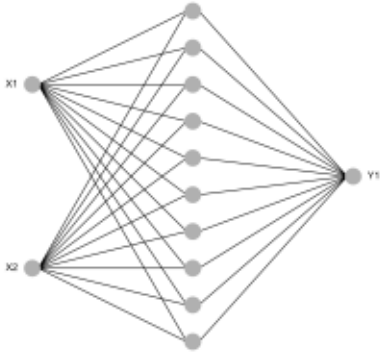
REGULARIZATION IN NONLINEAR MODELS

Setting: Classification for the spirals data. Neural network with single hidden layer containing 10 neurons, regularized with $L2$.
Varying λ affects smoothness of the decision boundary and magnitude of network weights:



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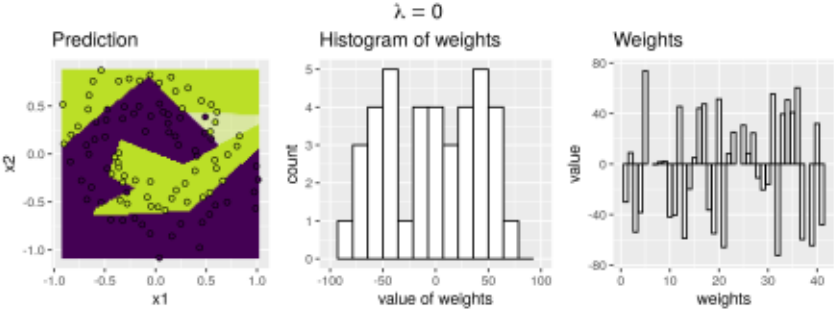

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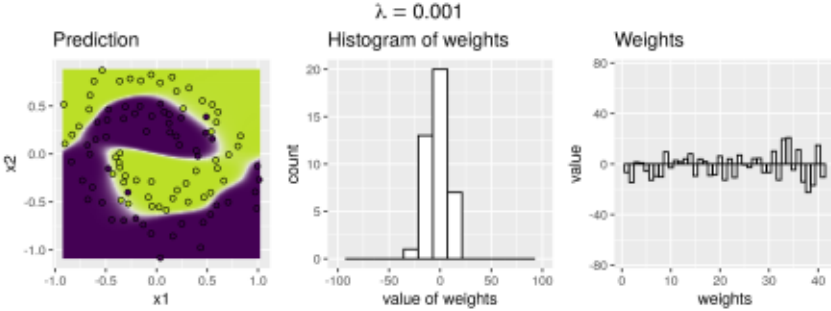

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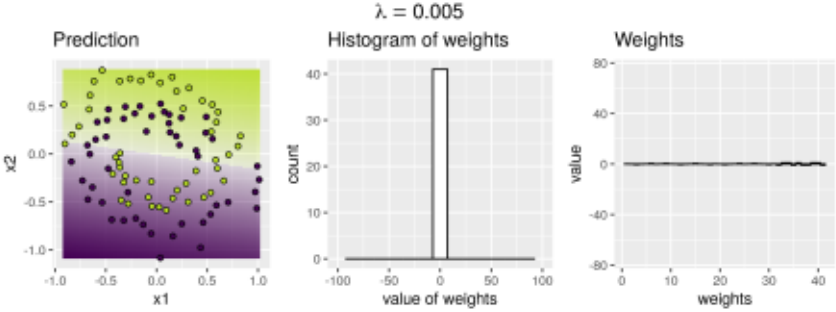

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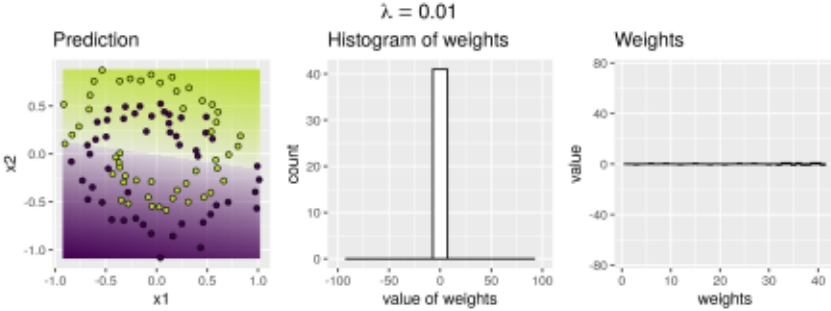
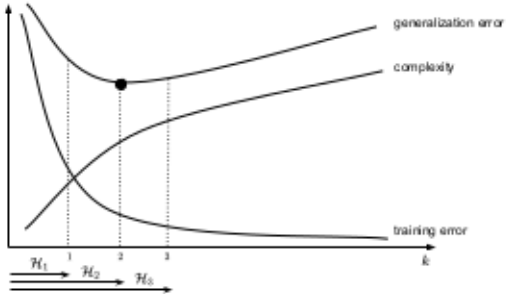


STRUCTURAL RISK MINIMIZATION / 2

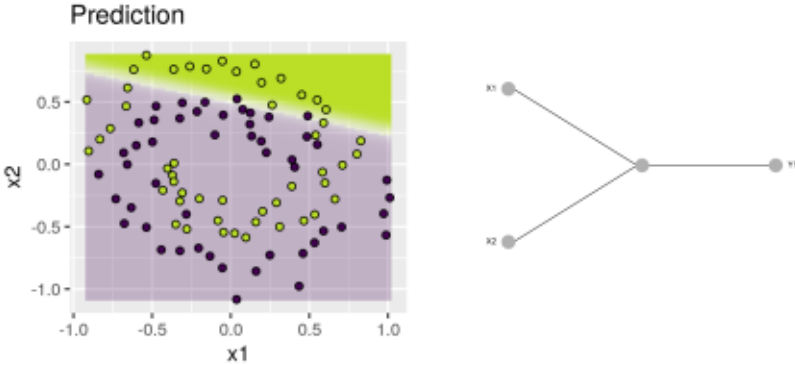
- SRM chooses the smallest k such that the optimal model from $\mathcal{H}_k$ found by ERM or RRM cannot significantly be outperformed by a model from a $\mathcal{H}_m$ with $m > k$.
- By this, the simplest model can be chosen, which minimizes the generalization bound.
- One challenge might be choosing an adequate complexity measure, as for some models, multiple complexity measures exist.



STRUCTURAL RISK MINIMIZATION

Setting: Classification for the spirals data. NN with 1 hidden layer and fixed (small) L2 penalty.
Varying the size of the hidden layer affects smoothness of the decision boundary:

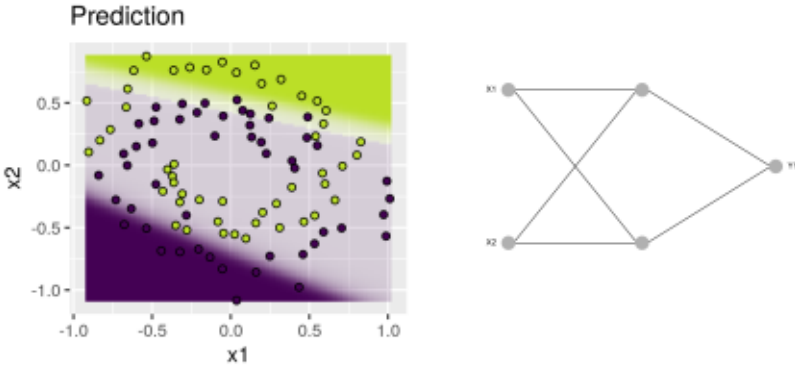
size of hidden layer = 1



STRUCTURAL RISK MINIMIZATION

Setting: Classification for the spirals data. NN with 1 hidden layer and fixed (small) L2 penalty.
Varying the size of the hidden layer affects smoothness of the decision boundary:

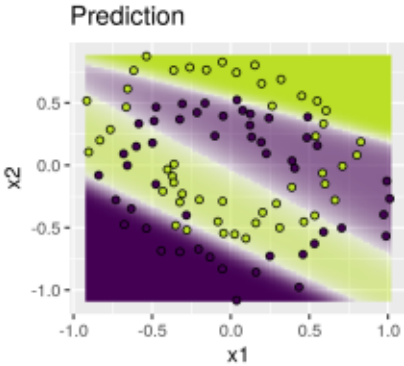
size of hidden layer = 2

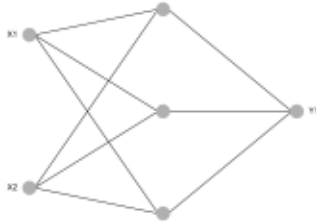


STRUCTURAL RISK MINIMIZATION

Setting: Classification for the spirals data. NN with 1 hidden layer and fixed (small) L2 penalty.
Varying the size of the hidden layer affects smoothness of the decision boundary:

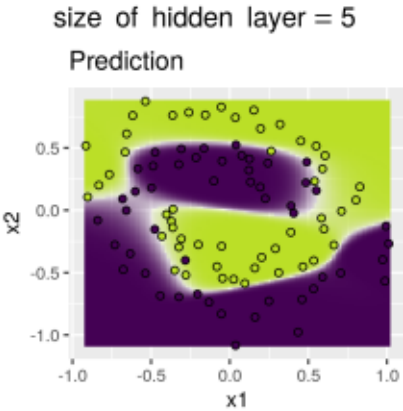
size of hidden layer = 3





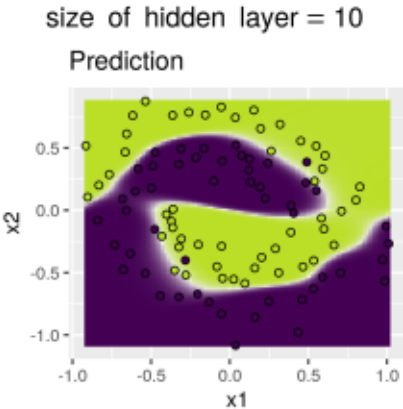
STRUCTURAL RISK MINIMIZATION

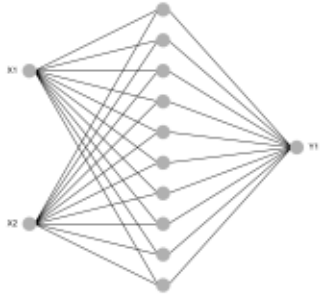
Setting: Classification for the spirals data. NN with 1 hidden layer and fixed (small) L2 penalty.
Varying the size of the hidden layer affects smoothness of the decision boundary:

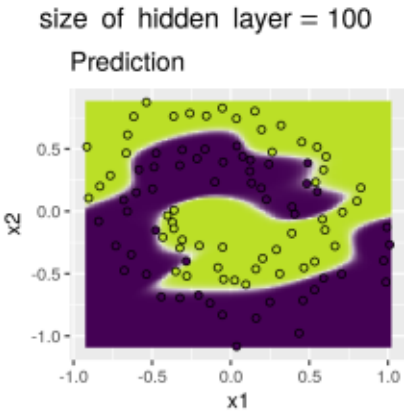
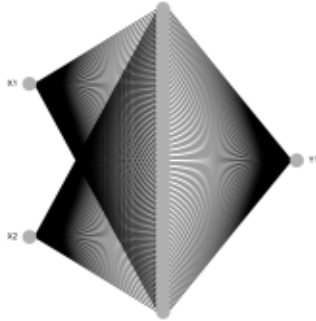


STRUCTURAL RISK MINIMIZATION

Setting: Classification for the spirals data. NN with 1 hidden layer and fixed (small) L2 penalty.
Varying the size of the hidden layer affects smoothness of the decision boundary:





STRUCTURAL RISK MINIMIZATION

Setting: Classification for the spirals data. NN with 1 hidden layer and fixed (small) L2 penalty.
Varying the size of the hidden layer affects smoothness of the decision boundary: